Robust comparison of multi-omics clustering algorithms for subtype discovery in chronic diseases

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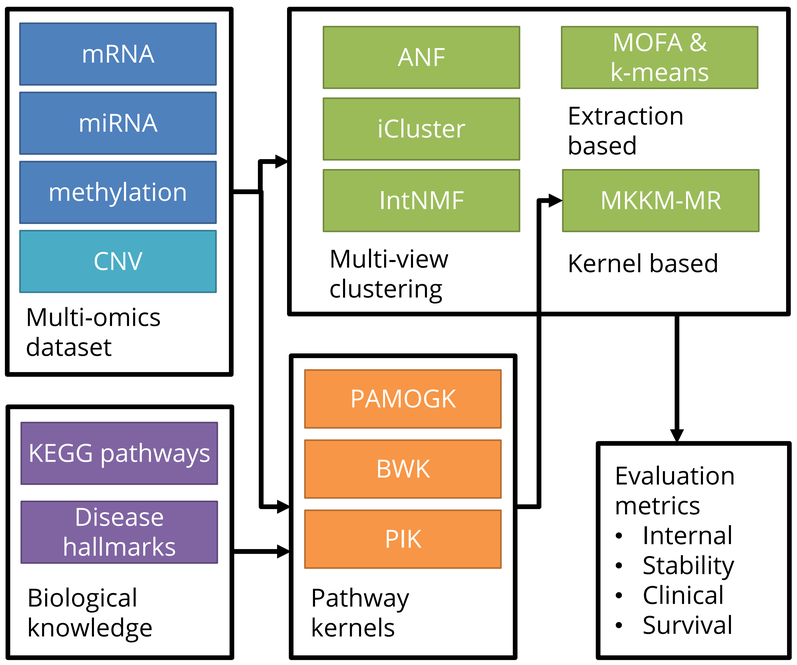
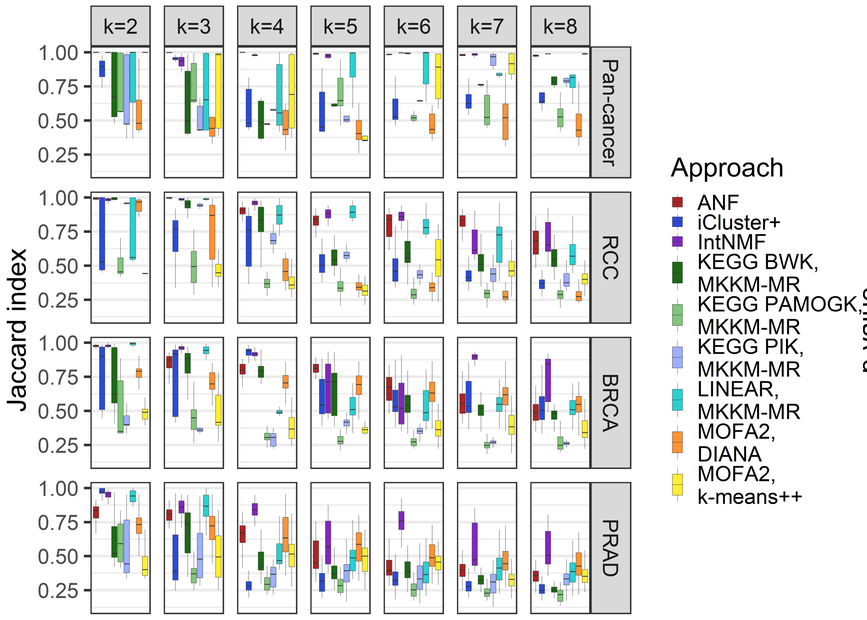
Introduction

Chronic diseases are highly complex and are driven by many different molecular mechanisms. Studying different molecular views or omics layers is imperative to discovering the exact mechanisms responsible for the development and progression of diseases. The progress in multi omics data collection via high throughput sequencing technologies has allowed clustering analysis spanning multiple omics layers. When comparing methods, studies tend to focus on clustering quality and known subtype agreement metrics as well as survival differences, but important covariates are not adequately considered. The reliability or reproducibility is also not considered. Our study aims to address these shortcomings by robustly testing popular methods in publicly available datasets.

Materials & Methods



We have implemented the COPS R-package to facilitate analysis of clustering stability and clinical relevance of (multi-)omics clustering approaches, that is available at github.com/vittoriofortino84/COPS. To test the algorithms in varying conditions, we performed case studies on four combinations of TCGA data, with one containing six different types of cancer (Pan-cancer), one containing three different histological types of renal cell carcinoma (RCC), one containing breast cancer (BRCA) samples and another containing prostate cancer (PRAD) samples. We considered mRNA and miRNA expression, and DNA methylation for clustering. For breast cancer we obtained better results after including copy number variations and excluding miRNAs. Mean methylation and miRNA across genes were computed via known associations.



Clustering stability in different datasets, number of clusters (k) and multi-view clustering algorithms.

Materials & Methods continued

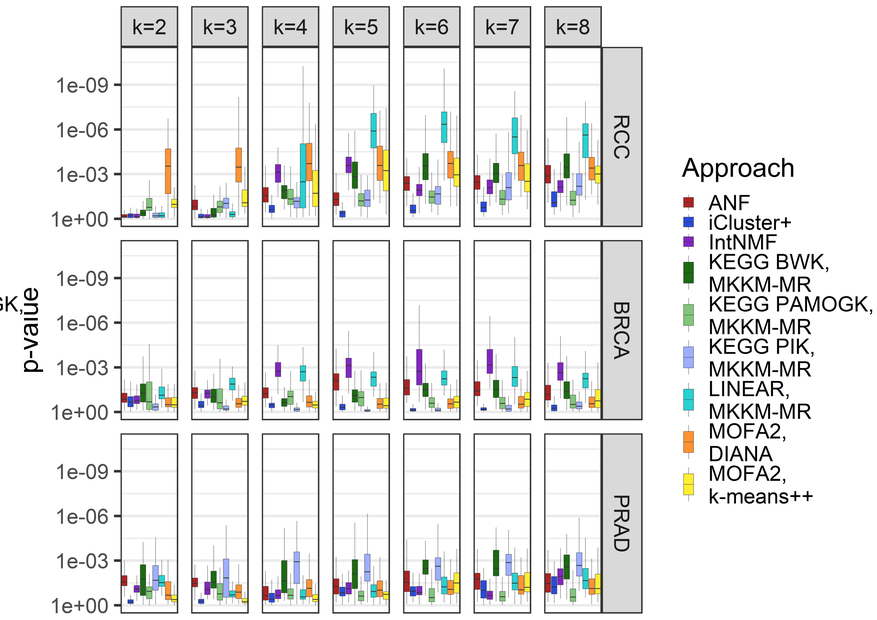
Stability was evaluated with the Jaccard index by comparing 10 times repeated 5-fold cross-validation results to a reference clustering obtained from the full dataset. Clinical relevance was evaluated with Cox proportional hazard models and the likelihood-ratio test. The Cox model was used to account for important covariates, specifically age and cancer stage, while the likelihood ratio test was used to estimate the significance of survival differences between clusters by using a model without clustering information as the baseline.

Results

For renal cell carcinoma, stability and survival relevance of the linear MKKM-MR, IntNMF, and ANF were notable with MKKM-MR yielding median p-value near 10^{-6} while staying above 0.8 stability for 5 clusters.

Case study details, number of samples and survival events

TCGA Datasets	Samples	Omics	Survival
BRCA, LGG, HNSC, THCA, LUAD, PRAD	3183	mRNA, miRNA, DNA methylation	Overall survival 615 events 2567 event-free
KIRC, KIRP, KICH	602	mRNA, miRNA, DNA methylation	Overall survival 129 events 473 event-free
BRCA	1075	mRNA, CNV, DNA methylation	Overall survival 148 events 927 event-free
PRAD	487	mRNA, miRNA, DNA methylation	Progression 90 events 397 event-free



Clinical relevance in different datasets, number of clusters (k) and multi-view clustering algorithms.

Results continued

For BRCA, IntNMF was the only method that yielded significant survival p-values and stability at the same time reaching p-values near 10^{-3} at 4 clusters at median stability near 0.9. For PRAD, most data-driven methods did not yield significant progression free survival, but one of the pathway kernel based methods, the pathway induced kernel (PIK), reached median p-values below 10^{-3} at 4 clusters, but the stability was very low at median below 0.5 at all number of clusters tested.

Discussion

The results indicate that current methods differ greatly in their potential for reproducible findings and ability to find actionable subtypes. While most algorithms consistently retrieved the six cancer types in the least challenging dataset, a few of them including iClusterPlus & MOFA struggled with poor stability beyond a few clusters. Integration of biological knowledge shows potential in some cases (Rintala et al. 2021), but the poor stability of the current kernel-based methods needs to be addressed. IntNMF showed good stability in high relevance results.

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